

Proposals should foresee appropriate resources to ensure close cooperation with the EC Knowledge Centre for Biodiversity (KCBD) and its Science Service.

2027

Consolidating biodiversity knowledge for nature and society

Proposals are invited against the following topic(s):

HORIZON-CL6-2027-01-BIODIV-01: Integrating Remote Sensing and in-situ observations of Biodiversity, towards a fully interoperable observation and data framework

Call: Call 01 - single stage (2027)	
Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 5.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 10.00 million.
<i>Type of Action</i>	Research and Innovation Actions
<i>Eligibility conditions</i>	The conditions are described in General Annex B. The following exceptions apply: The Joint Research Centre (JRC) may participate as member of the consortium selected for funding as a beneficiary with zero funding, or as an associated partner. The JRC will not participate in the preparation and submission of the proposal - see General Annex B. If projects use satellite-based earth observation, positioning, navigation and/or related timing data and services, beneficiaries must make use of Copernicus and/or Galileo/EGNOS (other data and services may additionally be used).
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply: Eligible costs will take the form of a lump sum as defined in the Decision of 7 July 2021 authorising the use of lump sum contributions under the Horizon Europe Programme – the Framework Programme for Research and Innovation (2021-2027) – and in actions under the

	Research and Training Programme of the European Atomic Energy Community (2021-2025) ¹¹² .
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Expected Outcome: Project results are expected to contribute to all of the following expected outcomes:

- advancing robust, policy-relevant ecosystem assessment, nature protection and restoration planning activities, and biodiversity trend prediction based on fit for purpose data, thus supporting EU biodiversity and climate objectives;
- strengthened capacity of researchers, practitioners and decision-makers to improve biodiversity monitoring practices and address knowledge gaps via the integration of data and observations across sensors and platforms, ranging from omics-based data (genomic, transcriptomic, metabolomic), various in-situ to satellite-based Earth observation data;
- enhanced usability of in-situ datasets as training and validation resources for statistical, machine learning and advanced AI-based approaches, in support of applications such as habitat classification, ecosystem mapping and biodiversity trend prediction.

Scope: Achieving the goals of the EU biodiversity strategy for 2030, the Kunming-Montreal Global Biodiversity Framework, and assessing progress towards their defined targets, requires coherent, integrated and long-term monitoring approaches, underpinned by FAIR (Findable, Accessible, Interoperable and Re-usable) data systems. Many existing in-situ data collections (i.e. genomic assessments, ground sampling, drone and airborne observations) were not designed with Satellite Remote Sensing integration in mind (i.e. for validation and calibration, integrated data products, or statistical scaling of information) and lack the structure, interface and metadata required to support advanced analytics, including AI. FAIR data on species and ecosystems will also help to ensure that biodiversity preservation is a mainstream feature of other sectors, such as agriculture, transport, energy or the bioeconomy. There is a need for systemic, harmonised or standardised biodiversity data at Earth's surface in order to support AI applications ranging from genome to space and to build up our knowledge on the status and trends of habitats, species, ecosystems, and on the drivers of decline.

To address these challenges, proposals should:

- develop and validate fit-for-purpose multisensory biodiversity data integration systems, to enable omics-based and in-situ data harmonization and integration with remote sensing data from space or sub-orbital platforms, as well as socio-ecological and climatic data for enhancing assessment of ecosystem condition and degradation, and the predictive modelling of biodiversity trends at international, regional and European scales;

¹¹² This [decision](https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf) is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

- develop concrete technical capabilities able to answer to diverse use cases, across terrestrial, freshwater and marine ecosystems in Member States and Associated Countries. To this end, proposals should identify and prioritise critical biodiversity knowledge gaps and their data needs, with a focus on predictive analytics and the use of AI, and address them by designing data integration approaches that support scientific and policy needs, including biodiversity protection, restoration, and sustainable use goals and targets, as well as other related policy objectives;
- develop demonstration and verification cases focusing on specific ecosystems under the EU ecosystem typology¹¹³ and habitats, including ecotones¹¹⁴, identified as those under pressure or as restoration priorities. Proposals should take into account socio-economic pressures and activities impacting ecosystems, including land and sea use and emissions of pollutants (e.g. agriculture, aquaculture, urbanization, resource exploitation and management practices). Demonstration cases should be relevant to EO applications, such as AI-assisted ecosystem and habitat mapping, ensuring interoperability with European and global frameworks such as the Global Ecosystem Typology as well as EUNIS, and aligned with reporting requirements under the Habitats Directive, Birds Directive, Marine Strategy Framework Directive and EU Nature Restoration Regulation;
- deliver FAIR data, harmonisation workflows and processing protocols supporting data integration from genome to space and verify the use of generated datasets on identified analysis needs. Proposals should link observation data to emerging socio-economic, climate, agriculture, forestry, fisheries and environmental data and to relevant data infrastructures, including, if applicable, the European Open Science Cloud (EOSC) and the European Common Data Spaces and SAGE¹¹⁵;
- contribute to consensus and implementation of joint definitions of data collection, metadata and processing protocols, data quality and harmonization of standards enabling the integration of observations “from genomes to space”.

Proposals should earmark resources for cooperation with existing initiatives such as in-situ data collection frameworks (e.g., GBIF, OBIS, LUCAS). The Joint Research Centre (JRC) may join the selected consortium in relation to effective coordination with the EC Knowledge Centre for Biodiversity (KCB) and its Science Service, as well as incorporating the in-situ data from European Commission steered biodiversity monitoring schemes such as LUCAS, specifically –LUCAS grassland, and EMBAL (European Monitoring of Biodiversity in

¹¹³ The “EU ecosystem typology” is a classification of ecosystem types developed by the Commission and the European Environmental Agency, in collaboration with Member States and it is tailored to meet EU needs. Its first level is legislated and is used for the reporting under in Annex IX of Regulation (EU) 691/2011.

¹¹⁴ Ecotones are areas of steep transition between ecological communities, ecosystems, and/or ecological regions along an environmental or other gradient. (S. Kark, Effects of Ecotones on Biodiversity, Reference Module in Life Sciences, Elsevier, 2017, ISBN 9780128096338, <https://doi.org/10.1016/B978-0-12-809633-8.02290-1>)

¹¹⁵ <https://www.egi.eu/project/sage/>.

Agricultural Landscapes). Proposals may consider services offered by European research infrastructures¹¹⁶.

This topic is part of the Biodiversity cluster of the EC-ESA Earth System Science Initiative (ESSI). Proposals should foresee sufficient mean for effective coordination with projects selected under ESA's FuturEO programme.

HORIZON-CL6-2027-01-BIODIV-02: Science-policy support to the implementation of EU and global biodiversity policies and strategies

Call: Call 01 - single stage (2027)	
Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 7.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 7.00 million.
<i>Type of Action</i>	Coordination and Support Actions
<i>Eligibility conditions</i>	The conditions are described in General Annex B. The following exceptions apply: The Joint Research Centre (JRC) may participate as member of the consortium selected for funding as a beneficiary with zero funding, or as an associated partner. The JRC will not participate in the preparation and submission of the proposal - see General Annex B.
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply: Eligible costs will take the form of a lump sum as defined in the Decision of 7 July 2021 authorising the use of lump sum contributions under the Horizon Europe Programme – the Framework Programme for Research and Innovation (2021-2027) – and in actions under the Research and Training Programme of the European Atomic Energy Community (2021-2025)¹¹⁷. Beneficiaries may provide financial support to third parties. The support to third parties can only be provided in the form of grants. The

¹¹⁶ The catalogue of European Strategy Forum on Research Infrastructures (ESFRI) research infrastructures portfolio can be browsed from ESFRI website <https://ri-portfolio.esfri.eu/>

¹¹⁷ This [decision](#) is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf